

Sugar And Flour—What's The Difference?

We've tested inkjet printers, and we've even told you about the kind of paper you should be using. Well, here's some information about the magical fluid that squirts from those tiny little nozzles!

Inks are of two types, Dye-based and Pigment based. The major difference between the two is the fact that dye-based inks are water-soluble, whereas pigment inks are non-water-soluble. Here the dye or pigment (as the case may be) is the colour component in the mixture with water. Water is, of course, the carrier of the colour that is either dissolved (dye) or suspended (pigment) in it.

Dye-based Inks

Think of a dye-based ink as a solution of sugar dissolved in water. Because there is no suspension involved, dye-based inks flow better, and are less likely to clog the minute nozzles on an inkjet printer. Because of their water solubility, any printout that has used dye-based inks will be more susceptible to damage when exposed to moisture. Another shortcoming of dye-based inks is their susceptibility to fading. Not only fade but also colour-shift, as we've noticed in some cases.

Dye-based inks work well on glossy paper, which doesn't completely absorb all the ink inside the paper, rather allowing some of it to actually dry on the surface—which is why the best photo results with dye-based inks are with glossy paper.

Pigment-based inks

In contrast to dye-based inks, pigment-based inks can be compared to a flour-in-water mixture. The flour doesn't dissolve; rather, it forms a suspension, also thickening the water slightly in the process. Due to this, pigment inks are thicker, and also tend to clog a printer's inkjet nozzles more easily. However, modern pigment-based inks have remedied this situation to a large extent, but the problem is of an inherent nature and therefore cannot be done away with entirely.

Pigmented inks also fade slower, owing to their larger molecular structure, and therefore ultimately thicker coating on the paper. Due to the fact that the pigment isn't water soluble, they are also water-resistant, though not completely waterproof. On exposure to water, 5 ~ 10 per cent of the colours may run, which is significantly lower than the damage that would be caused in a similar situation with a dye-based ink.

There are disadvantages to pigment-based inks as well. They're costlier than dye inks. Also, dye inks give much better colours, and much sharper, more vivid printouts. This is the reason why most manufacturers use pigment inks in their black cartridges, but prefer to use dye inks in their colour cartridges. Incidentally, there is a lot of research on to produce cheaper hybrid inks, which will have the benefits of both ink types.

HP Deskjet D2360

Quality on a budget

HP's D2360 is a compact-looking printer sporting a shiny pure-white top combined with a matte-finished grey body. Needless to say, it'll draw looks atop any desk. Build quality is good per se, though the D2360 does exude that built-to-a-budget feel.

As with most HP printers, the input tray doubles as the output tray—a neat design. Ergonomics is a sore spot—you need to lift the top cover to open the input/output tray. Another sore point is the bulky power adapter that HP supplies—all the other non-HP printers had power adapters built in.

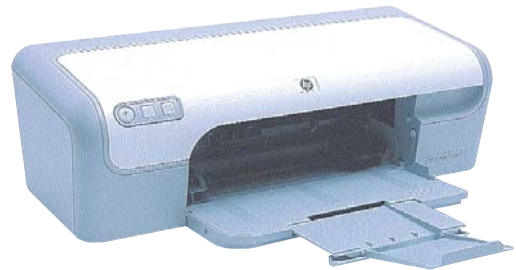
One word that describes HP's driver interface? Excellent! The settings are very transparent, and virtually all the adjustments can be done from a single page. As with the D4168, you

can create multiple profiles for each type of print job—monochrome, text/image, photo printing, and more.

HP doesn't use individual ink cartridges, and makes do with the regular two tanks. While not so noticeable on this printer, its bigger sibling—the D4168 demonstrates how individual tanks aren't always necessary for good photograph printing.

Surprisingly, the D2360 has some issues with text printing. Some aliasing was noticeable, and the smaller fonts weren't as well defined as we'd have liked on paper.

While miles ahead of Epson's C79 and inches ahead of the Canon iP1700, the D2360 does have something to boast about. If only we'd tested it before we did the D4168, we'd have been happier too!



The combination test was a close affair, especially at lower settings. Once you set your printer to guzzle ink (the higher settings), both the Canon iP1700 and Epson's C79 leave the D2360 coughing in their dust trails. The D2360 does score in the highlighting-on-the-fruit department, something both Epson and Canon miss out on.

Our test photo saw very decent colour reproduction on the whole. Specular highlights, once unheard of on inkjets, is now very much a reality, and HP doesn't miss out here.

While the D2360 languishes in the shadow of its more expensive sibling, it makes for an attractive alternative for home use.

At just Rs 2,999, it does put the competition to shame. ■

michael_browne@thinkdigit.com

Contact Sheet		Inkjet Printers	
Brand	Company	Phone	E-mail
Canon	Canon India Pvt Ltd	1800-345-33-66	info@canon.co.in
Epson	Epson India Pvt Ltd	1800 425 0011	murthy@eid.epson.co.in
HP	HP India Pvt Ltd	124 - 2566 111	vibhor.bansal@hp.com